

Daniel He

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Education

Western University - BSc., Honours Specialization in Computer Science *September 2024 - May 2028 Expected*
With Admission Scholarship, Scholarship of Distinction (\$2500) | **GPA: 3.50/4.00**

Work Experience

OpsGuru (AWS Premier Partner) - Software Engineering Intern *May 2026 - August 2026*
AWS AI/ML Infrastructure & Agent Systems

- Built a reusable **AWS Bedrock AgentCore** accelerator with 9 AI agents using Python, **LangGraph**, and Strands, integrating Bedrock Knowledge Bases, S3, DynamoDB, and **Terraform** infrastructure used by **15 engineers**
- Designed and developed a Deep Agent accelerator with multi-step planning and token-aware memory management, reducing retained memory tokens by **~40%** through threshold-based compression

Western University - Undergraduate Research Assistant *October 2025 - May 2026*
LLM Hallucination - Dr. Apurva Narayan

- Co-authored *HalluciBench*, an LLM hallucination benchmarking paper, under review at ICLR 2027
- Designed and implemented an LLM-as-a-judge benchmarking pipeline in Python with **Ollama** to evaluate hallucination rates across **29 LLMs**, incorporating Hyperparameter Tuning and **Prompt Engineering**
- Implemented a **Reinforcement Learning**-inspired **Synthetic Data** generation pipeline from a research paper to create a **1000-sample** dataset for evaluating LLM refusal calibration and hallucination rates

Scotiabank - Software Engineering Intern *May 2025 - August 2025*
IT&S Capital Markets/Data Analytics

- Refactored a broken **Airflow** DAG pipeline to store vector embeddings in **ChromaDB** with persistence to **Google Cloud Storage**, migrating embedding logic into a Cloud Function. Cut p95 latency by **~90%** and resolved a 6-month production issue, enabling a scalable RAG knowledge base for the bank's internal knowledge bot
- Enhanced internal research summarization pipeline by extending PDF parsing support from 2 to 4 document formats, automated deployment using **CI/CD** build configuration tools, and documented using Jira and Confluence
- Built a **Python** cyclomatic complexity analyzer using SonarQube and Selenium, reducing analysis time by **~85%**

Projects

Dex2 ([GitHub Link](#), [Video-Demo](#)) *February 2026*

- Built a **TypeScript/React.js** Chrome extension that auto-captures page content to trigger context-aware agent actions (edit slides, open tabs, draft emails) using **MongoDB** for embeddings and Supabase (**PostgreSQL**)
- Implemented AI **Agent Orchestration** using FastAPI for hybrid retrieval and **Tool Calling** with background workers and concurrency to keep chat non-blocking. Used **LangChain** for chunking and retrieval

Mini-v ([GitHub Link](#)) *April 2026*

- Built a **C++** LLM inference server with micro-batching and continuous batching, achieving **2.5x throughput** improvement (6.5 to 16.8 req/s) across concurrent load with 100% request success rate

Say Less ([Devpost Link](#), [Live-Demo](#)) *November 2025*

Hack Trent 2025 - MLH Best Use of ElevenLabs (21 total submissions)

- Created a website that converts American Sign Language (ASL) to speech and speech to text in real time
- Spearheaded development of a **FastAPI** backend with **RESTful API** endpoints, integrating a fine tuned **MediaPipe** model (**~80%** accuracy on ASL to text). Containerized with **Docker** and deployed on **Render**

Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL | Spoken Languages: English, Chinese (Mandarin)
Frameworks/Libraries: Next.js, React.js, FastAPI, MediaPipe, Tensorflow, PyTorch, SQLAlchemy, LangChain, LangGraph
Developer Tools: Docker, Git, Vercel, GCP, Apache Airflow, AWS, Ollama, Linux, Claude Code, Cursor, Terraform
Databases: PostgreSQL (Supabase), SQLite, MongoDB, ChromaDB